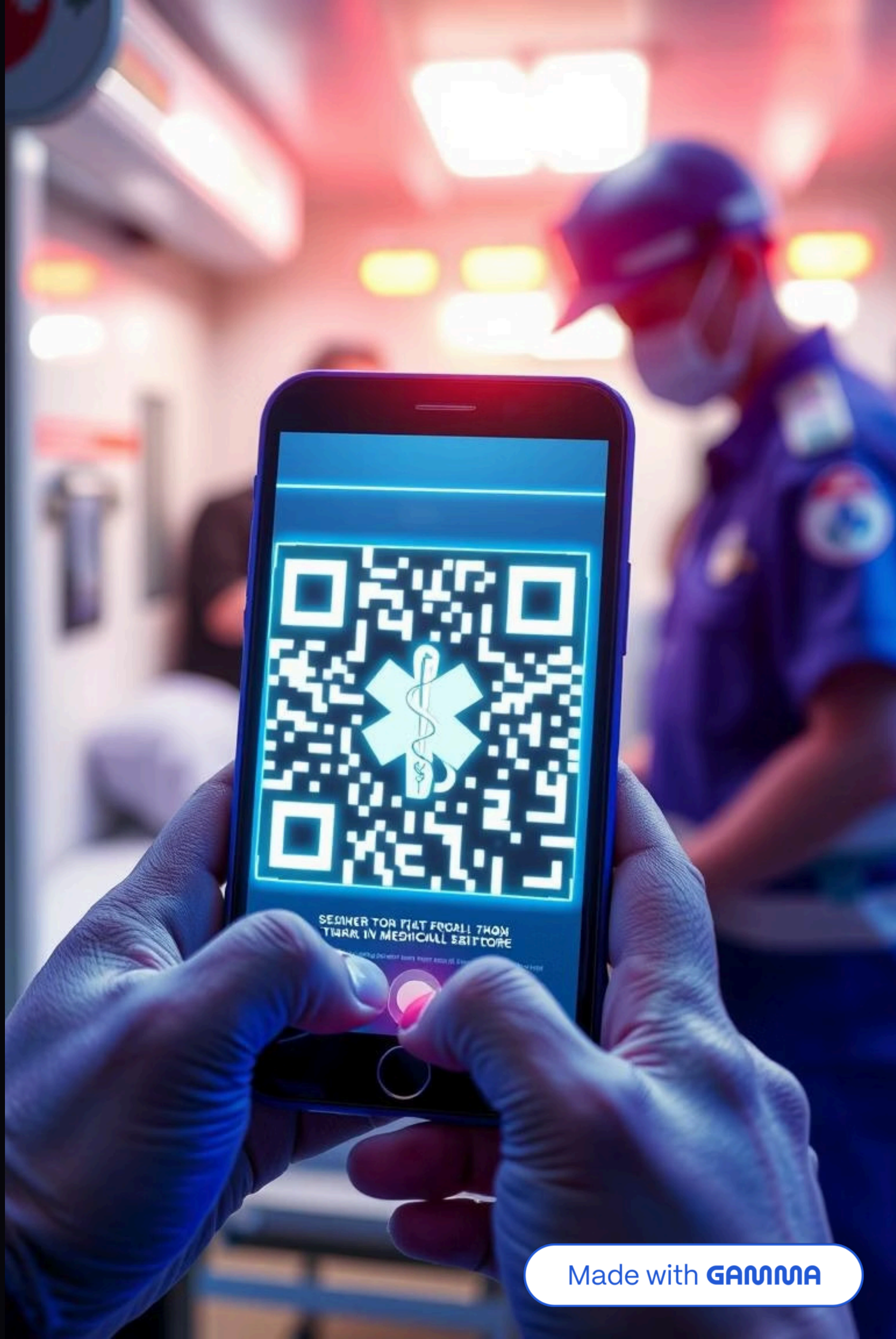
Lifeline-QR

Welcome to the Future of Emergency Care



Meet the Project Team

Introducing the dedicated individuals driving the Lifeline QR project forward.

Salah Mohy Mohamed

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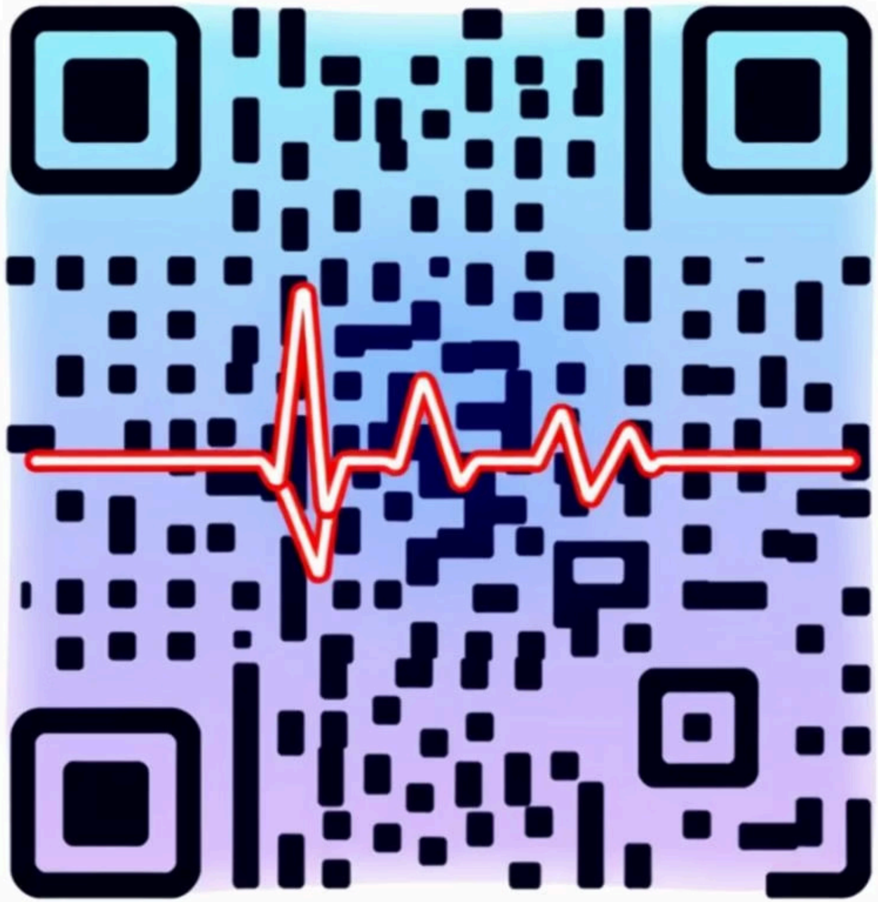
Mohamed Ashraf Sayed Abdullah

Background

Technological advancements in the healthcare field have greatly improved the quality of medical services and the ability to save lives. Among these innovations comes the idea of using a QR code that contains a patient’s medical history and laboratory results, allowing healthcare providers to access this information instantly in emergency situations or accidents.



Lifeline QR -System for Emergency Medical Data Access

Problem Statement

During emergencies or when a patient is unconscious, doctors often face difficulties in identifying the patient’s medical history — such as chronic diseases, allergies, or previous test results — which can delay treatment or lead to inaccurate medical decisions.

Objective

1. To develop a smart system that uses a QR code to store essential medical data for each patient.
2. To allow medical teams to instantly access these records by scanning the QR code.
3. To reduce medical errors and improve emergency response time.

Benefits

1

Saves patients' lives

by providing immediate access to crucial medical information.

2

Reduces the risk

of medical errors and mistreatment.

3

Enhances communication

between hospitals and laboratories.

4

Keeps a digital, secure, and easily accessible record

of patients' medical history.

Scope

The project initially focuses on developing a web site and a database system for hospitals within one governorate (e.g., Minya), with the potential to expand to a national level. The system includes:

- Creating a digital medical file for every patient.
- Linking each file with a unique QR code.
- Allowing authorized medical personnel to update and upload new test results and reports.
- Implementing access permissions to ensure data privacy and security

BMC MODEL

Lifeline QR Business Model Canvas

Our business model outlines how Lifeline QR creates, delivers, and captures value. It details key aspects from partnerships to revenue.

Key Partners

Hospitals, clinics, labs, insurance, Ministry of Health, web hosts, cybersecurity firms.

Key Activities

Website development, database management, QR code generation, healthcare partnerships, data protection.

Key Resources

Secure servers, database system, web team, medical consultants, QR tools.

Value Proposition

Instant, secure access to patient medical history via QR code for emergency responders.

Customer Relationships

Continuous data updates, 24/7 support, privacy assurance, user-friendly interface.

Channels

Official website, hospital partnerships, online registration, health campaigns, social media.

Customer Segments

Patients, hospitals, emergency teams, insurance companies, health ministries.

Cost Structure

Development, hosting, security, maintenance, marketing, technical support.

Revenue Streams

Subscriptions for hospitals, premium accounts, data management contracts, sponsorships.

Digital Business Model Canvas

LifeLine QR

Key Partners

Hospitals and clinics – Medical laboratories – Health insurance companies – Ministry of Health – Web hosting providers – Cybersecurity firms

Key Activities

Developing and maintaining the website – Managing a secure medical database – Generating and linking QR codes – Partnering with healthcare institutions – Ensuring data protection and privacy

Key Resources

Secure web servers – Database management system – Web development team – Medical consultants – QR code generation tools

Value Proposition

A web-based platform that stores and provides instant access to patients' medical history through a QR code, allowing doctors and emergency responders to retrieve vital health data quickly and securely.

Customer Relationships

Continuous medical data updates – 24/7 website support – Privacy assurance – User-friendly interface for both patients and medical staff

Channels

Official website – Hospital partnerships – Online registration – Health awareness campaigns – Social media outreach

Customer Segments

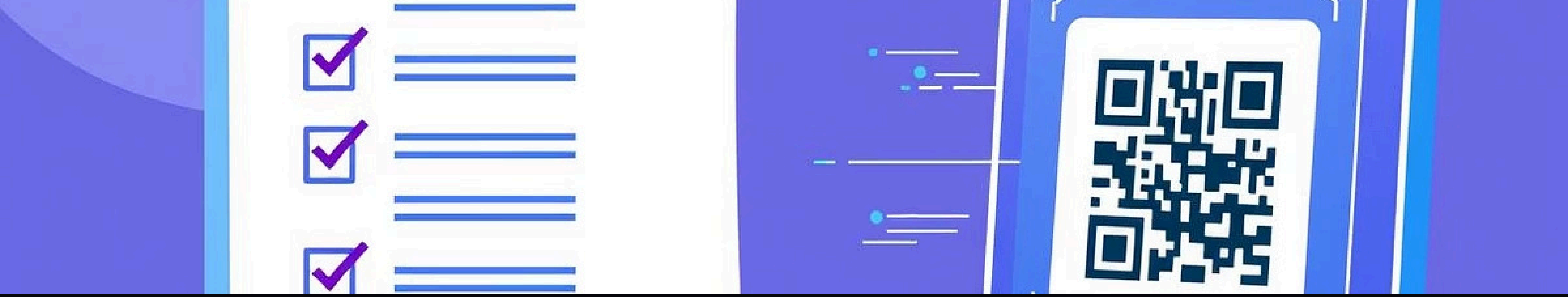
Individual patients – Hospitals and clinics – Emergency response teams – Insurance companies – Health ministries

Cost Structure

Website development and hosting – Database security – Maintenance and updates – Marketing and awareness – Technical support

Revenue Streams

Subscription plans for hospitals and clinics – Individual premium accounts – Data management contracts with health organizations – Sponsorships or government funding



Functional Requirements

1. The system must allow the creation of a patient file with a unique patient ID.

2. The system must generate a unique QR code for each patient.

3. The system must allow doctors to scan the QR code to access patient data.

4. The system must display the patient's medical history immediately after scanning.

5. The system must store medical records including diseases, allergies, and medications.

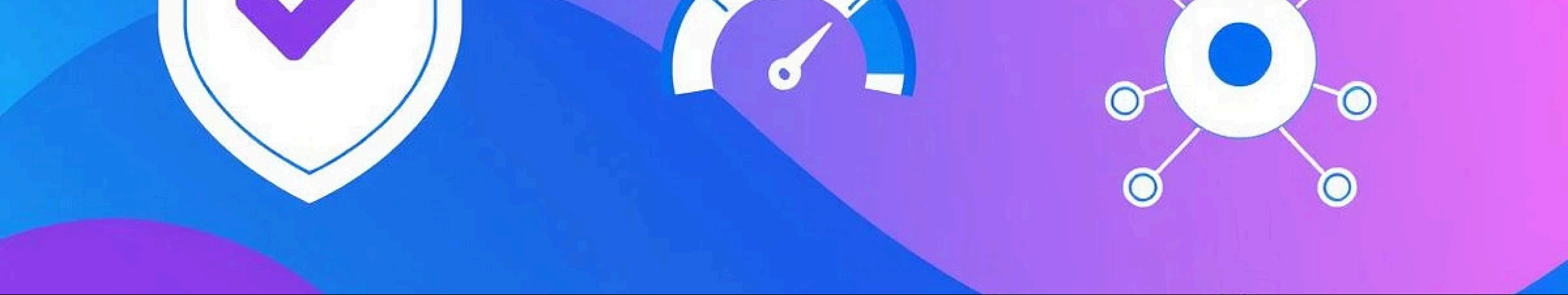
6. The system must allow adding test results to the medical record.

7. The system must log every access made by a doctor to patient data.

8. The system must restrict access to authorized doctors only.

9. The system must allow updating medical records when needed.

10. The system must retain historical test results for each patient.



Non-Functional Requirements

🔐 Security

- The system must ensure the confidentiality of patient data.
- Only authorized doctors should be able to access patient information.
- All access attempts must be logged.
- Data must be protected from unauthorized modification.

⚡ Performance

- The system must display patient data within seconds after scanning the QR code.
- The system must handle multiple access requests without delay.

🌐 Availability

- The system must be available 24/7, especially for emergencies.
- The system must minimize downtime.

📈 Scalability

- The system must support an increasing number of patients and doctors.
- The database must efficiently handle growing medical records.

🔁 Reliability

- The system must provide accurate and consistent medical data.
- The system must prevent data loss in case of system failure.

👤 Usability

- The system interface must be simple and easy to use.
- Medical staff must be able to use the system without special training.

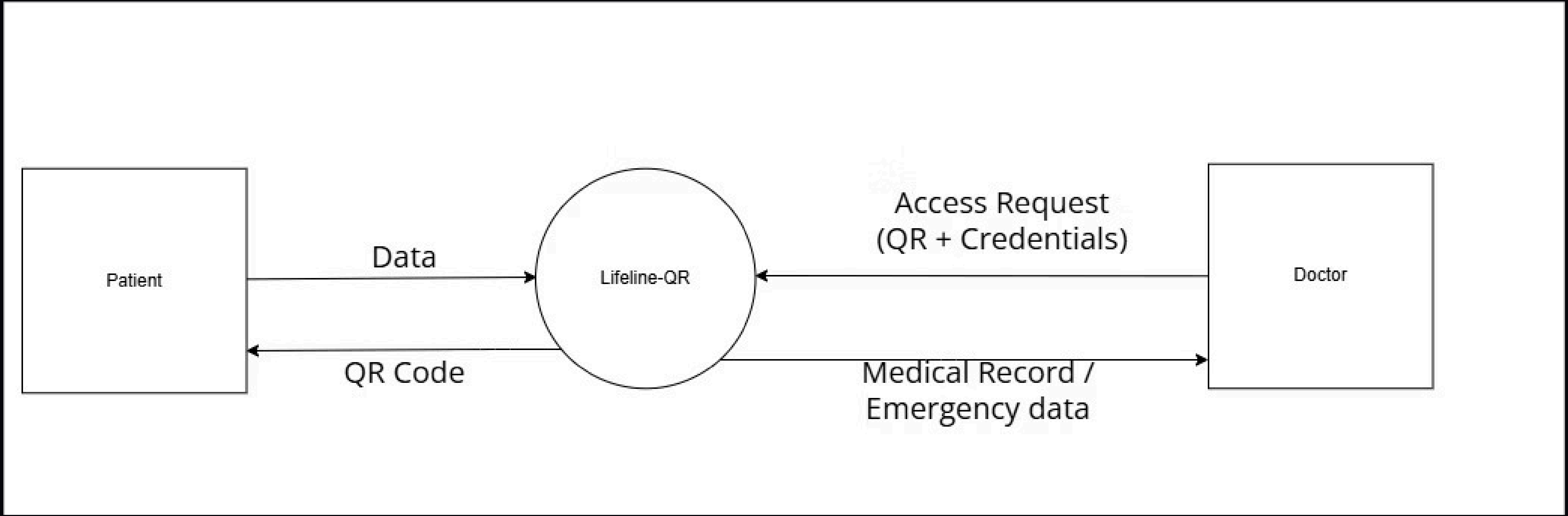
🛠️ Maintainability

- The system must allow for easy updates and modifications.
- Errors must be easy to detect and fix.

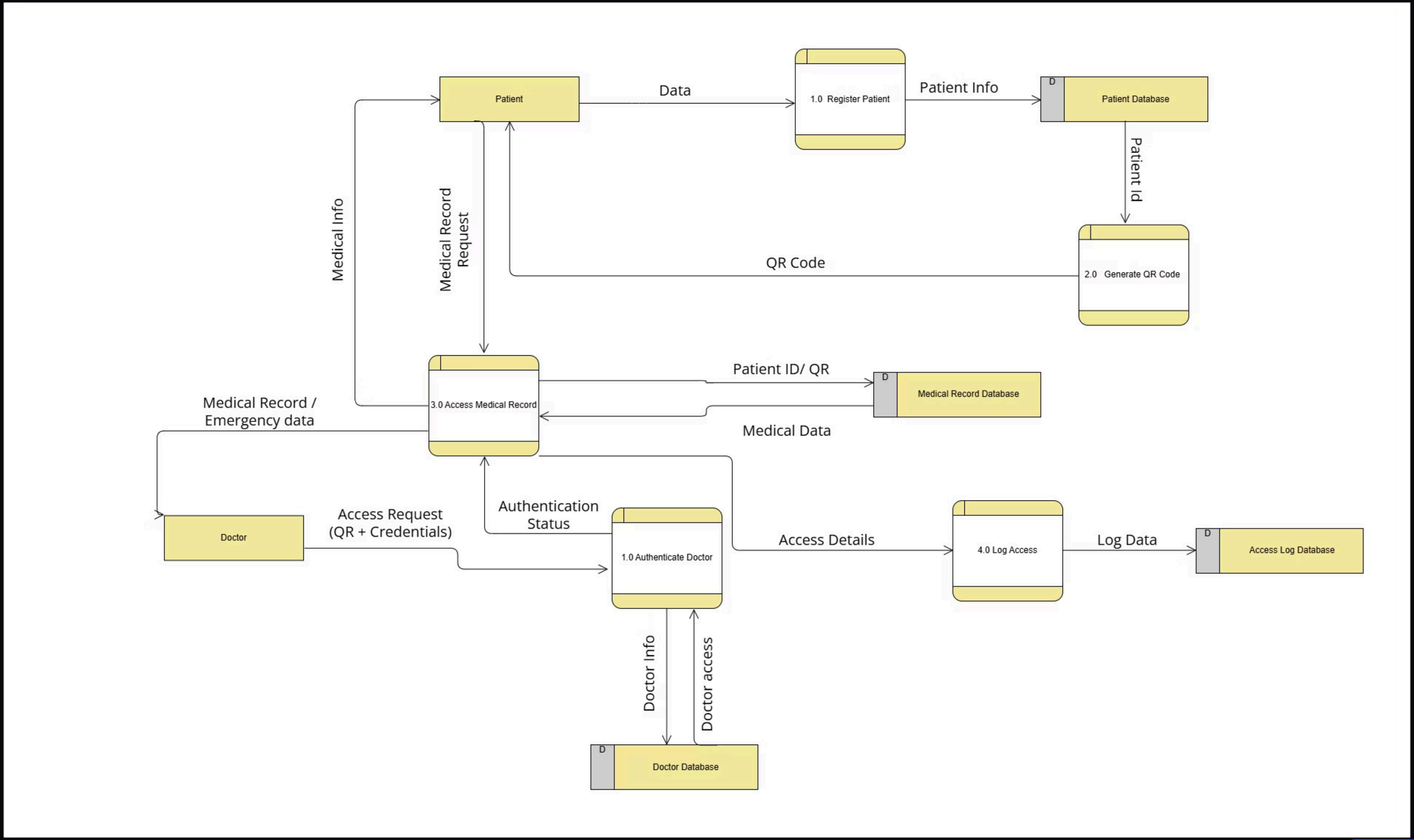
🔗 Interoperability

- The system must be able to integrate with hospital systems.
- The system must support standard medical data formats.

CONTEXT DIAGRAM

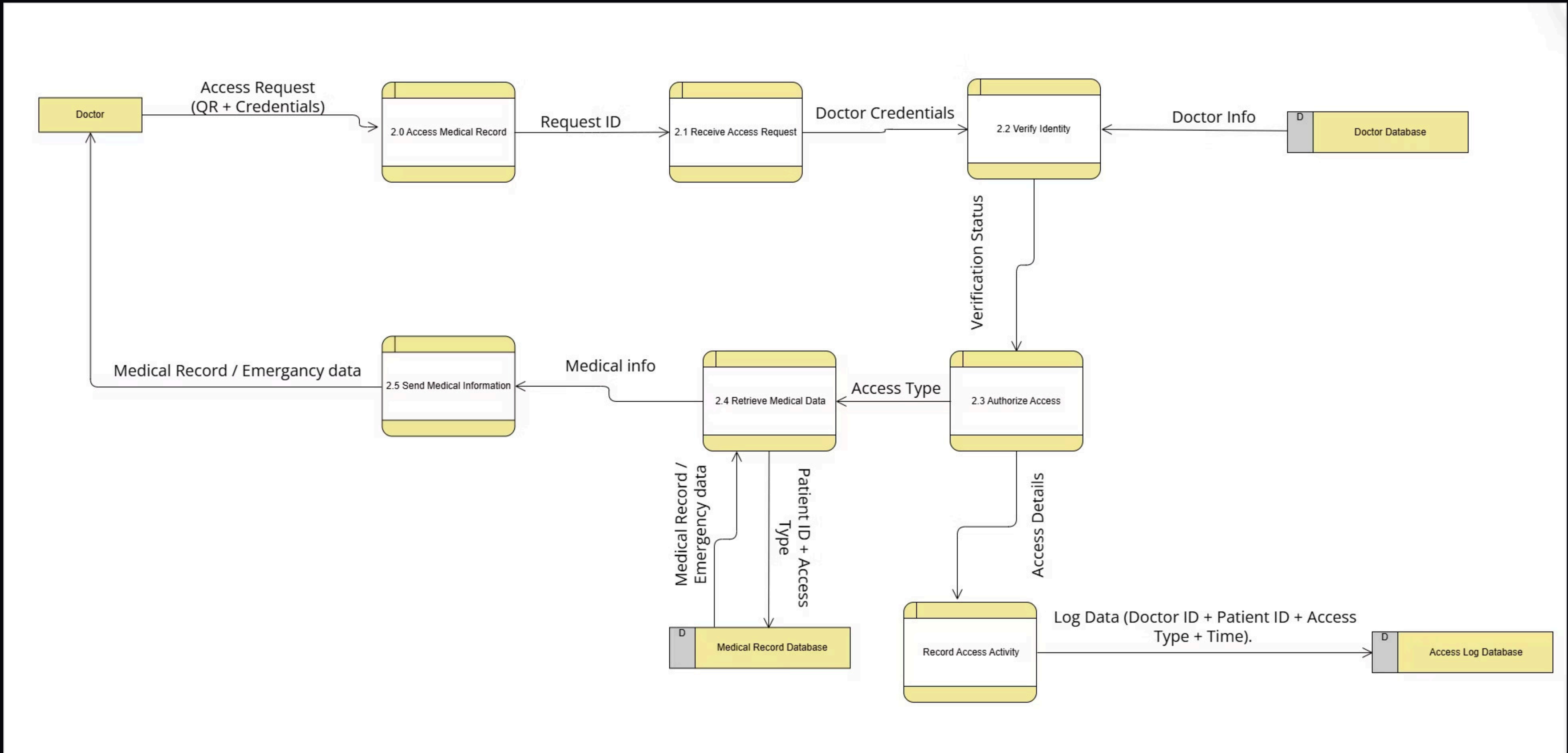


DFD LEVEL 0



Patient
Data
1.0 Register Patient
Patient Info
D
Patient Database
Patient Id
2.0 Generate QR Code
QR Code
Medical Info
Medical Record Request
3.0 Access Medical Record
Patient ID/ QR
D
Medical Record Database
Medical Data
Medical Record / Emergency data
Doctor
Access Request (QR + Credentials)
Authentication Status
1.0 Authenticate Doctor
Access Details
4.0 Log Access
Log Data
D
Access Log Database
Doctor Info
Doctor access
D
Doctor Database

DFD LEVEL 1



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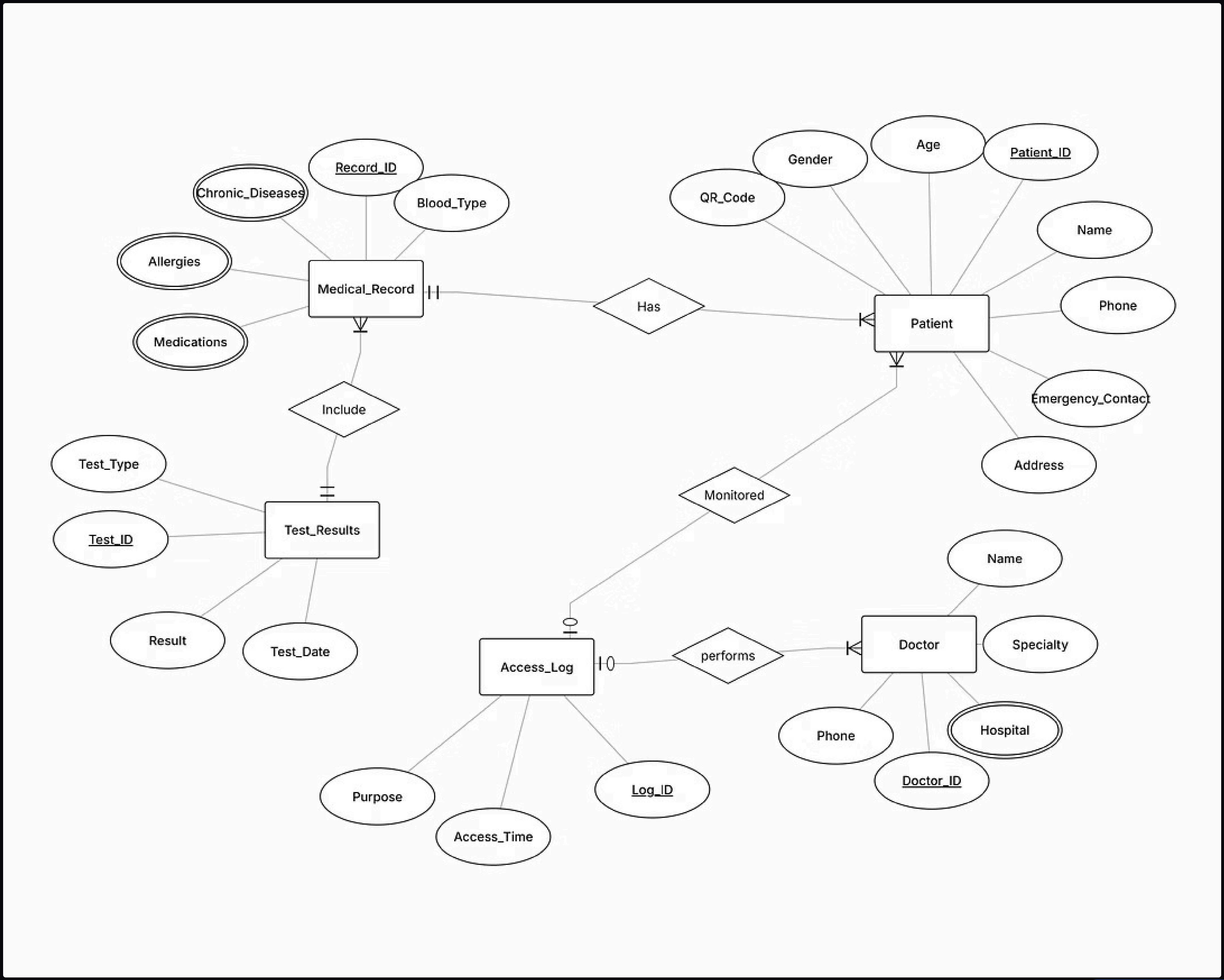

Record_ID
Chronic_Diseases
Blood_Type
Allergies
Medical_Record
Medications
Include
Test_Type
Test_ID
Test_Results
Result
Test_Date
Has
Gender
Age
Patient_ID
QR_Code
Name
Phone
Patient
Emergency_Contact
Address
Monitored
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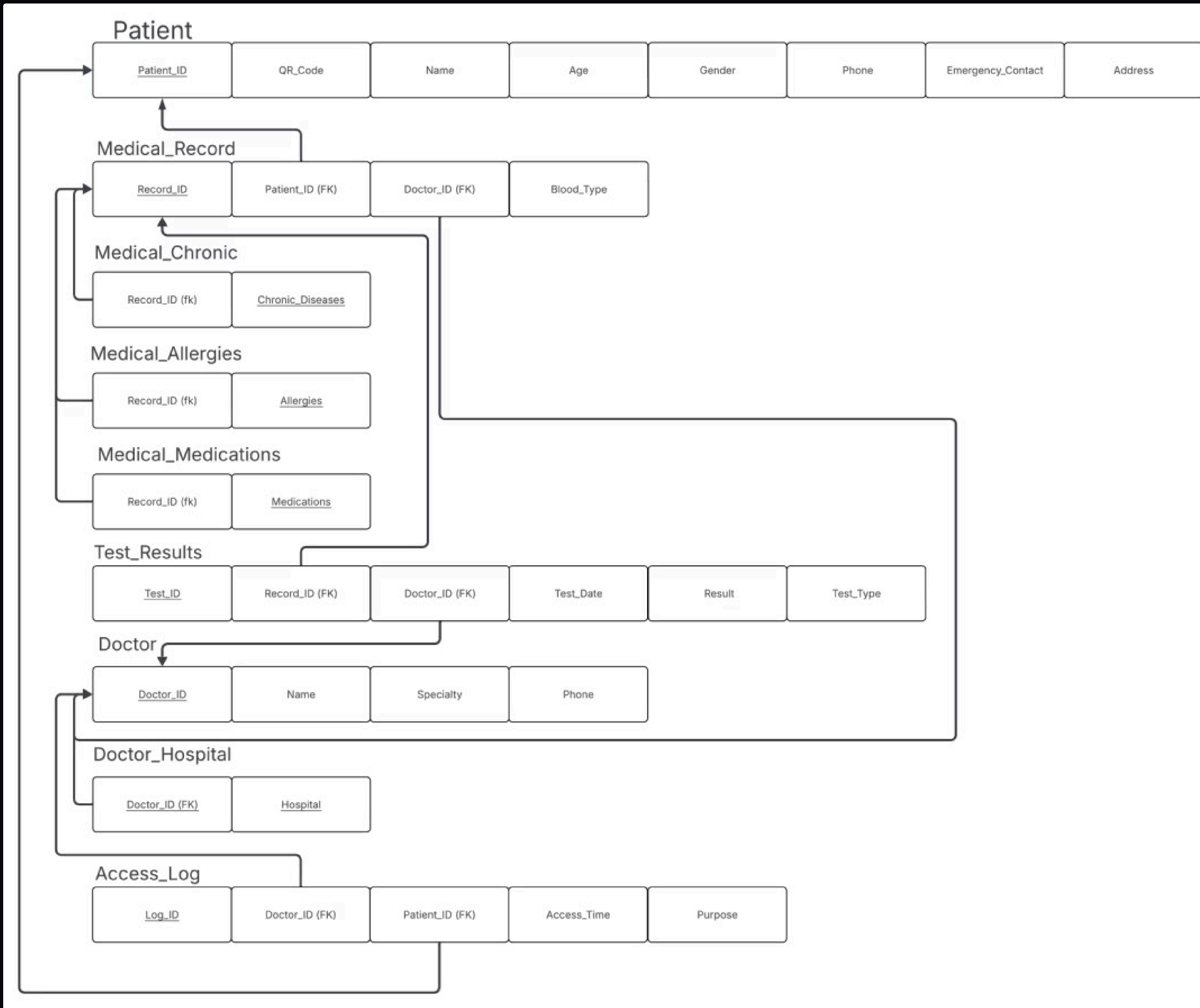
Access_Log
performs
Doctor
Specialty
Phone
Hospital
Doctor_ID
Purpose
Access_Time
Log_ID

SCHEMA


Patient
Patient_ID
QR_Code
Name
Age
Gender
Phone
Emergency_Contact
Address
Medical_Record
Record_ID
Patient_ID (FK)
Doctor_ID (FK)
Blood_Type
Medical_Chronic
Record_ID (fk)
Chronic_Diseases
Medical_Allergies
Record_ID (fk)
Allergies
Medical_Medications
Record_ID (fk)
Medications
Test_Results
Test_ID
Record_ID (FK)
Doctor_ID (FK)
Test_Date
Result
Test_Type
Doctor
Doctor_ID
Name
Specialty
Phone
Doctor_Hospital
Doctor_ID (FK)
Hospital
Access_Log
Log_ID
Doctor_ID (FK)
Patient_ID (FK)
Access_Time
Purpose